

Smart City Training Laboratory

Assignment Brief

Task Information

Task Title: Smart Parking System with Plate Recognition

Mode: Smart City Simulation

Environment: Parking Area

Due Date: *(Set by instructor)*

Maximum Score: 100 Marks

Objective

Design and simulate a smart parking system that can detect parking slot usage, identify vehicles using a camera, control entry through an automated barrier, and integrate a microservice to handle vehicle verification and receipt generation.

Learning Outcomes

By completing this assignment, you will be able to:

- Configure and connect smart city components
- Use a Smart City Gateway to control system behavior
- Simulate real-world system workflows
- Integrate microservices into a simulation
- Understand how multiple subsystems interact in a smart environment

Task Description

You are required to build a smart parking system that:

- Detects whether parking slots are free or occupied
- Captures vehicle plate numbers at entry
- Verifies vehicles using a service
- Controls a barrier to allow or deny entry
- Tracks parking usage
- Generates a receipt after parking

Requirements

Simulation Setup

Use the **Parking Area environment** and add:

- At least **7 Vehicle Sensors** (for parking slots)
- **1 Parking Camera** (for plate detection)
- **1 Smart Barrier** (entry control)
- **1 Smart City Gateway** (system controller)

Connections

You must:

- Connect all vehicle sensors to gateway inputs (I1–I7)
- Connect the camera to CAM1
- Connect the barrier to output O1
- Connect at least one microservice using an API port

Gateway Configuration

You must write logic to:

- Count available parking spaces
- Check if a vehicle is allowed
- Open or close the barrier
- Trigger system actions based on conditions

Microservice Integration

Create at least one microservice, such as:

- Plate Verification Service
- Parking Session Service
- Receipt Generation Service

The service should:

- Receive input data
- Process the request
- Return a response

System Workflow

Vehicle arrives at parking entrance
↓
Camera captures plate number
↓
Gateway sends plate to verification service
↓
If vehicle is allowed and space is available
→ Barrier opens
↓
Vehicle enters and occupies a slot
↓
System updates available parking spaces
↓
When vehicle exits
→ Generate receipt

Deliverables

You must submit:

- A completed simulation project
- Configured Smart City Gateway logic
- At least one working microservice
- A functioning system (entry → parking → exit)

Marking Criteria

Criteria	Marks
Component setup	20
Connections	20
Gateway logic	20
Microservice integration	20
System functionality	20
Total	100

Instructions

- Ensure all components are correctly connected

- Test your system before submitting
- Follow the workflow requirements
- Submit your project before the deadline

This is **ready to paste into your system** as:

Task Document / Assignment File / PDF Content

If you want next, I can:

- Generate a **second assignment (Smart Traffic with AI cameras)**
- Or create **student submission report template**
- Or design **grading UI for the trainer**